

Cauchy Stress Tensor

Derivation:

- 0) Start from Newton's second law
 - 1) Show surface forces dominant $V \rightarrow 0$
 - 2) Cauchy postulate (local)
 - 3) Law of action & reaction (Newton's 3rd law)
 - 4) Cauchy's theorem
- $\Rightarrow$ Cauchy stress

1) Force balance as $V \rightarrow 0$

2nd law: $\underline{f} = m \underline{a}$

resultant forces: $\underline{f} = \underline{f}_b + \underline{f}_s$

$\Rightarrow \underline{f}_s = m \underline{a} - \underline{f}_b$

mass: $m = \int_{\Omega} \rho dV$

body force: $\underline{f}_b = \int_{\Omega} \underline{b} dV$

surface force: $\underline{f}_s = \oint_{\partial\Omega} \underline{t} dA$



V_{Ω} = volume

$A_{\partial\Omega}$ = area

Limit of vanishing body

$$\lim_{\delta \rightarrow 0} \underline{\tau}_s = m \underline{a} - \underline{\tau}_b ?$$

mass: $\lim_{\delta \rightarrow 0} m = \int_{\Omega} \rho(\underline{x}) dV \approx \rho(\underline{x}_0) \int_{\Omega} dV = \rho_0 \underline{V}_{\Omega}$

body force: $\lim_{\delta \rightarrow 0} \underline{\tau}_b = \int_{\Omega} \rho g dV \approx \rho(\underline{x}_0) g \int_{\Omega} dV = \rho_0 g \underline{V}_{\Omega}$

substitute into 2nd law

$$\lim_{\delta \rightarrow 0} \oint_{\partial\Omega} \underline{\tau} dA = \int_{\Omega} \rho \underline{a} - \rho g dV \approx \rho_0 (\underline{a} - g) \underline{V}_{\Omega}$$

normalize by surface area!

$$\lim_{\delta \rightarrow 0} \frac{1}{A_{\partial\Omega}} \oint_{\partial\Omega} \underline{\tau} dA = \frac{\underline{V}_{\Omega}}{A_{\partial\Omega}} \rho_0 (\underline{a} - g)$$

Volume vanishes faster than surface area!

$$\lim_{\delta \rightarrow 0} \frac{\underline{V}_{\Omega}}{A_{\partial\Omega}} = 0$$

Consider a sphere: $\underline{V}_{\Omega} = \frac{4}{3} \pi \delta^3$ $A_{\partial\Omega} = 4 \pi \delta^2$

$$\lim_{\delta \rightarrow 0} \frac{\underline{V}_{\Omega}}{A_{\partial\Omega}} = \frac{\delta}{3} = 0 \quad (\text{holds for any shape})$$

Surface forces vanish of infinitesimal body

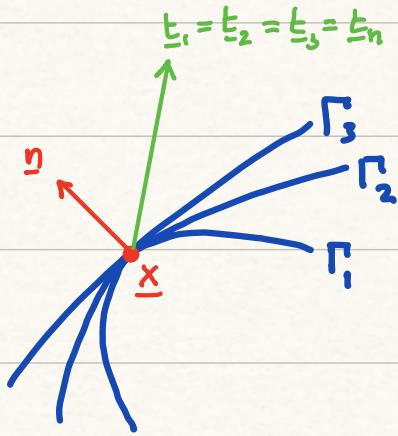
$$\lim_{\delta \rightarrow 0} \frac{1}{A_\Omega} \oint_{\partial\Omega} \underline{t} \, dA = 0$$

Note:

- $\frac{1}{A_\Omega}$ normalization
- assume ρ , $|g|$ and $|a|$ are finite

$\Rightarrow$ basis for derivation of Cauchy stress tensor.

2) Cauchy's Postulate



Traction depends point wise
on normal $\Rightarrow$ local

$$\underline{t} = \underline{t}(\underline{n}(\underline{x}), \underline{x})$$

Note: there are non-local theories!

$$\underline{t} = \underline{t}(\underline{n}, \nabla \underline{n}, \underline{x}) \quad \text{advanced stuff}$$

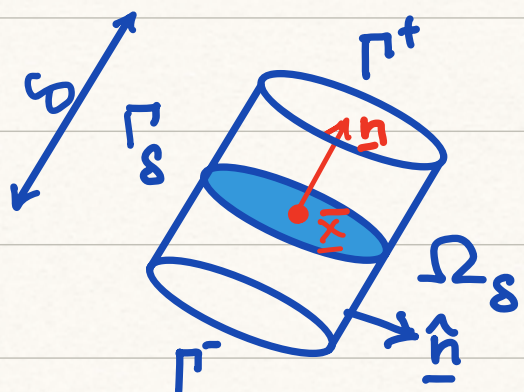
$\uparrow$
curvature of Γ

3. Law of Action and Reaction

$$\underline{t}(-\underline{n}, \underline{x}) = -\underline{t}(\underline{n}, \underline{x})$$

$\underline{t}$ is continuous & bounded

Proof: Force balance on cylinder



$$\partial\Omega_\delta = \Gamma^+ \cup \Gamma^- \cup \Gamma_\delta$$

$$\text{on } \Gamma^+: \underline{\hat{n}} = \underline{n}$$

$$\text{on } \Gamma^-: \underline{\hat{n}} = -\underline{n}$$

$$\Gamma^\pm = D \quad \text{area of disk}$$

$$\lim_{\delta \rightarrow 0} \Gamma_\delta = 0$$

Surface force:

$$\underline{f}_s = \int_{\partial\Omega_\delta} \underline{t}(\underline{n}, \underline{x}) dA = \underline{0}$$

$$\underline{f}_s = \int_{\Gamma_\delta} \underline{t}(\underline{\hat{n}}, \underline{x}) dA + \int_{\Gamma^+} \underline{t}(\underline{n}, \underline{x}) dA + \int_{\Gamma^-} \underline{t}(-\underline{n}, \underline{x}) dA = \underline{0}$$

$$\lim_{\delta \rightarrow 0} \int_{\Gamma_\delta} \underline{t}(\underline{\hat{n}}, \underline{x}) dA + \underbrace{\int_{\Gamma^+} \underline{t}(\underline{n}, \underline{x}) dA}_{D \underline{t}(\underline{n}, \underline{\bar{x}})} + \underbrace{\int_{\Gamma^-} \underline{t}(-\underline{n}, \underline{x}) dA}_{D \underline{t}(-\underline{n}, \underline{\bar{x}})} = 0$$

$$\Rightarrow \underline{t}(\underline{n}, \underline{\bar{x}}) + \underline{t}(-\underline{n}, \underline{\bar{x}}) = \underline{0}$$



Cauchy's Theorem

If $\underline{t}(\underline{n}, \underline{x})$ satisfies Cauchy's postulate
then $\underline{t}(\underline{n}, \underline{x})$ is linear in $\underline{n}$

$$\Rightarrow \underline{t}(\underline{n}, \underline{x}) = \underline{\underline{\sigma}}(\underline{x}) \underline{n} \quad \underline{\underline{\sigma}}(\underline{x}) \text{ Cauchy stress}$$

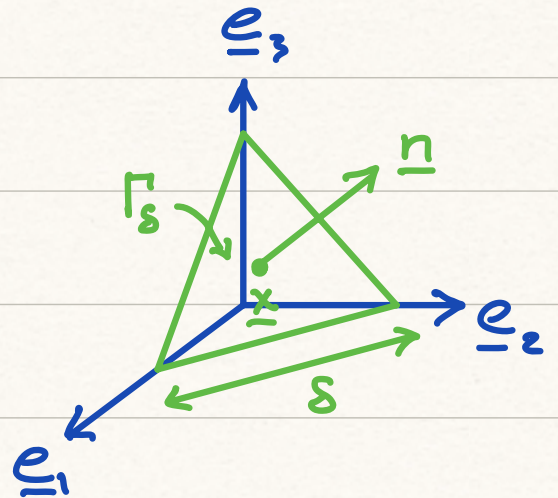
Proof:

Ω_δ tetrahedron

$$\partial\Omega_\delta = \Gamma_\delta \cup \Gamma_1 \cup \Gamma_2 \cup \Gamma_3$$

$$\Gamma_\delta: \underline{n} \cdot \underline{e}_i > 0.$$

$$\Gamma_i: \underline{n}_j = -\underline{e}_j$$

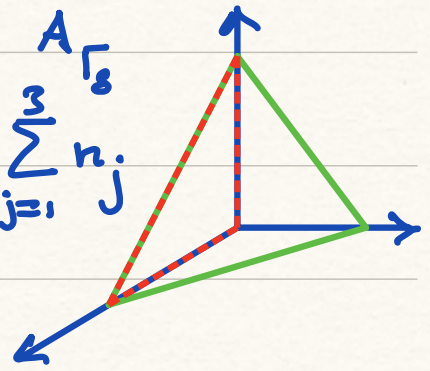


$$\lim_{\delta \rightarrow 0} \frac{1}{A_{\partial\Omega_\delta}} \int_{\partial\Omega_\delta} \underline{t}(\underline{n}(\underline{y}), \underline{y}) dA = 0$$

$$\lim_{\delta \rightarrow 0} \frac{1}{A_{\partial\Omega_\delta}} \left[\int_{\Gamma_\delta} \underline{t}(\underline{n}, \underline{y}) dA + \sum_{j=1}^3 \int_{\Gamma_j} \underline{t}(-\underline{e}_j, \underline{y}) dA \right] = 0$$

Homework 1 you showed: $A_{\Gamma_j} = n_j A_{\Gamma_s}$

$$A_{\partial\Omega_s} = A_{\Gamma_s} + \sum_{j=1}^3 A_{\Gamma_j} = \lambda A_{\Gamma_s} \quad \lambda = 1 + \sum_{j=1}^3 n_j$$



substituting

$$\lim_{\delta \rightarrow 0} \frac{1}{A_{\partial\Omega_s}} \left[\int_{\Gamma_s} \underline{t}(\underline{n}, \underline{x}) dA + \sum_{j=1}^3 \int_{\Gamma_j} \underline{t}(\underline{n}(-\underline{e}_j, \underline{y})) n_j dA \right] = \underline{0}$$

$$\lim_{\delta \rightarrow 0} \frac{1}{\lambda A_{\Gamma_s}} \underbrace{\int_{\Gamma_s} \underline{t}(\underline{n}, \underline{y}) + \sum_{j=1}^3 \underline{t}(-\underline{e}_j, \underline{y}) n_j dA}_{f(\underline{y})} = \underline{0}$$

$$\lim_{\delta \rightarrow 0} \frac{1}{\lambda A_{\Gamma_s}} \int_{\Gamma_s} f(\underline{y}) dA = \frac{1}{\lambda A_{\Gamma_s}} f(\underline{x}) A_{\Gamma_s} = \frac{1}{\lambda} f(\underline{x}) = 0 \Rightarrow f(\underline{x}) = 0$$

$$\Rightarrow \underline{t}(\underline{n}, \underline{x}) + \sum_{j=1}^3 \underline{t}(-\underline{e}_j, \underline{x}) n_j = \underline{0}$$

$$\underline{t}(\underline{n}, \underline{x}) = - \sum_{j=1}^3 \underline{t}(-\underline{e}_j, \underline{x}) n_j$$

using action & reaction:

$$\underline{t}(\underline{n}, \underline{x}) = \sum_{j=1}^3 \underline{t}(\underline{e}_j, \underline{x}) n_j$$

traction is weighted sum of tractions on coord. planes

$$\begin{aligned}\underline{t}(\underline{n}, \underline{x}) &= \underline{t}(\underline{e}_1, \underline{x}) n_1 + \underline{t}(\underline{e}_2, \underline{x}) n_2 + \underline{t}(\underline{e}_3, \underline{x}) n_3 \\ &= \underbrace{[\underline{t}(\underline{e}_1, \underline{x}), \underline{t}(\underline{e}_2, \underline{x}), \underline{t}(\underline{e}_3, \underline{x})]}_{\underline{\sigma}(\underline{x})} \underbrace{\begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}}_{\underline{n}}\end{aligned}$$

$$\Rightarrow \underline{t}(\underline{n}, \underline{x}) = \underline{\sigma}(\underline{x}) \underline{n}$$

We can also use dyadic property

$$\underline{t}(\underline{n}, \underline{x}) = \underline{t}(\underline{e}_j, \underline{x}) n_j \quad \text{summation convention}$$

compare

$$\begin{aligned}(\underline{t}(\underline{e}_j, \underline{x}) \otimes \underline{e}_j) \underline{n} &= (\underline{e}_j \cdot \underline{n}) \underline{t}(\underline{e}_j, \underline{x}) & \underline{n} &= n_i \underline{e}_i \\ &= n_i (\underline{e}_j \cdot \underline{e}_i) \underline{t}(\underline{e}_j, \underline{x}) \\ &= \underline{t}(\underline{e}_j, \underline{x}) n_j\end{aligned}$$

$$\begin{aligned}\Rightarrow \underline{t}(\underline{n}, \underline{x}) &= (\underline{t}(\underline{e}_j, \underline{x}) \otimes \underline{e}_j) \underline{n} & \underline{t} &= t_i \underline{e}_i \\ &= (t_i (\underline{e}_j, \underline{x}) \underline{e}_i \otimes \underline{e}_j) \underline{n}\end{aligned}$$

$$\underline{t}(\underline{n}, \underline{x}) = \underline{\sigma}(\underline{x}) \underline{n}$$

$$\underline{\sigma} = \sigma_{ij} \underline{e}_i \otimes \underline{e}_j \quad \text{with} \quad \sigma_{ij} = t_i(\underline{e}_j, \underline{x})$$

Hence σ_{ij} is the i -th component of the traction on the j -th coordinate plane.

The traction vectors on the coord. planes at $\underline{x}$ are

$$\underline{t}(\underline{e}_1, \underline{x}) = t_i(\underline{e}_1, \underline{x}) \underline{e}_i = \sigma_{i1}(\underline{x}) \underline{e}_i$$

$$\underline{t}(\underline{e}_2, \underline{x}) = t_i(\underline{e}_2, \underline{x}) \underline{e}_i = \sigma_{i2}(\underline{x}) \underline{e}_i$$

$$\underline{t}(\underline{e}_3, \underline{x}) = t_i(\underline{e}_3, \underline{x}) \underline{e}_i = \sigma_{i3}(\underline{x}) \underline{e}_i$$

